Market Data Stream Topology Proposal

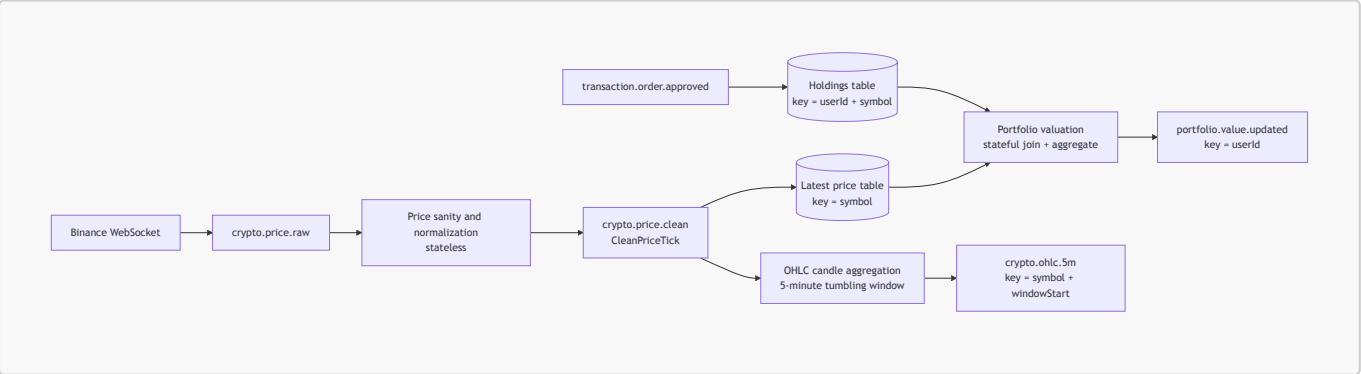
Status: proposal
Purpose: define one coherent stream-processing topology that demonstrates stateless, stateful, and windowed event processing using the existing CryptoFlow market-data stream.

Summary

Use the existing Binance ingestion in `market-data-service` as the source. It already publishes raw price ticks to `crypto.price.raw`. Add a Kafka Streams topology that derives three business streams from this input:

Concept	Processor	Input	Output
Stateless	Price sanity and normalization	`crypto.price.raw`	`crypto.price.clean`
Stateful	Real-time portfolio valuation	`crypto.price.clean`, `transaction.order.approved`	`portfolio.value.updated`
Windowed	5-minute OHLC candles	`crypto.price.clean`	`crypto.ohlc.5m`

Flowchart



Processor Responsibilities

1. Price Sanity and Normalization

This is the stateless stage. Each price tick is handled independently.

Responsibilities:

- Drop malformed events with missing symbol, price, or timestamp.
- Drop impossible prices, for example `price <= 0`.
- Normalize symbol representation, for example consistently using `BTCUSDT` or `BTC-USDT`.
- Add stream metadata such as `sourceVenue = "binance"` and `processedAt`.

- Convert the current JSON event into an Avro-backed `CleanPriceTick` event at the `crypto.price.clean` boundary.

This centralizes checks that are currently scattered across ordinary Spring Kafka consumers.

2. Real-Time Portfolio Valuation

This is the stateful stage. It continuously maintains derived state from multiple event streams.

Responsibilities:

- Materialize the latest price per symbol from `crypto.price.clean`.
- Materialize user holdings from `transaction.order.approved`.
- Join holdings with latest prices.
- Compute position values as `quantity * latestPrice`.
- Aggregate positions by `userId`.
- Emit `portfolio.value.updated` whenever a relevant price or holding changes.

This is not portfolio validation. It is continuous portfolio valuation. The value depends on remembered state: latest prices and current holdings.

3. OHLC Candle Aggregation

This is the windowed stage. It turns the unbounded price stream into finite 5-minute time buckets.

Responsibilities:

- Group clean ticks by symbol.
- Use **event time** from the price tick timestamp via a custom `TimestampExtractor` reading `sourceTimestamp` from the payload (see `docs/agent-instructions/events/32-timestamp-extractor.md`). Lecture 10 is explicit that the active time semantics are decided here, not by broker config alone.
- Apply a 5-minute tumbling window.
- Compute open, high, low, close, and tick count for each symbol/window.
- **Suppress** until the window closes so each `(symbol, window)` emits exactly one closed candle (matches trading-style "candle close" semantics; see scope 5 and `33-suppress-operator.md`).
- Emit candles to `crypto.ohlc.5m`.

Late events should be accepted within a small grace period, for example 10–30 seconds. Events later than the grace period can be dropped or routed to a late-event topic.

Implementation Notes

- Keep `crypto.price.raw` as the replayable source topic.
- Treat `crypto.price.clean` as the canonical downstream price contract.
- Prefer Avro for new derived topics: `crypto.price.clean`, `portfolio.value.updated`, and `crypto.ohlc.5m`.
- The existing in-memory `LocalPriceCache` and REST-time valuation can remain during migration, but the stream topology should become the demonstrated stateful implementation.

- A dedicated streams module or service is preferable to embedding the topology inside form/onboarding workflows.

Time semantics across the topology

Drawn from Lecture 10:

- **Stage 1 (sanity)** — stateless; time semantics are not load-bearing here, but the stage MUST preserve the source `sourceTimestamp` field on `CleanPriceTick` so that downstream stages can use payload-driven event time.
- **Stage 2 (valuation)** — stateful, non-windowed; outputs inherit the latest contributing input timestamp. No grace period to choose, but Kafka Streams will still process records in **timestamp order** across the price and order inputs (time-driven data flow), which is what makes valuation deterministic.
- **Stage 3 (OHLC)** — windowed; pinned to event time via a custom `TimestampExtractor` and closed via grace + suppress as above.

A consistent custom extractor across stages 2 and 3 is the cheapest way to make sure reprocessing produces identical valuations and identical candles.